



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Faculty of Science and Technology

Department of Computer Science

CSC 3312: SOFTWARE ENGINEERING

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Project Title

AgriAssist – A Simple Mobile App to Support Farmers

Supervised By

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Submitted By

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1. INTRODUCTION

1.1 Background to the Problem

Agriculture is one of the most important sectors in Bangladesh. Farmers play a vital role in both food production and national economy. However, many farmers face challenges in managing their agricultural activities efficiently.

In most cases, farmers do not have access to all necessary farming information in one place. Information about weather conditions, crop market prices, fertilizer usage, irrigation methods and crop diseases is often scattered and difficult to access. As a result, farmers may make incorrect or delayed decisions, which can lead to crop damage and financial loss.

In rural areas, internet connectivity is sometimes limited, making it harder to access online information when needed. Although smartphone usage is increasing, there is still no single integrated system that provides complete agricultural support services within one platform.

The proposed system, AgriAssist, is designed to help address these challenges. It is a mobile-based application that includes Farmer's Help, Smart Irrigation Support, Crop Disease Finder, Voice Assistant, Offline Mode, and Farming Calendar. The system aims to assist farmers in managing their daily activities more effectively and improving overall productivity.

Target Group of Users:

Farmer: The primary users of the AgriAssist system are Farmers. They can access accurate weather forecasts, monitor crop market prices, and receive reminders for planting, watering, fertilizing and harvesting, helping them manage their activities more effectively.

Farm laborer: Farm laborers can follow clear instructions for daily tasks, which reduces errors and improves efficiency.

Agricultural advisor: Agricultural advisors can provide guidance more effectively and monitor crop health with ease. They can also share updated information with the farmers.

System administrator: System administrators will maintain system data, manage user accounts ensuring that the application runs properly. The system is designed to support all the users so that they can work together in one platform and help farmers in carrying out their daily agricultural activities in a more organized and efficient manner.

1.2 Selection of Process Model

Scrum model will be best for AgriAssist project.

SCRUM: Scrum is an Agile framework that helps teams deliver projects in short, iterative cycles called sprints (usually 2–4 weeks). Work is organized into a product backlog (a prioritized list of features), and at the start of each sprint the team selects tasks for the sprint backlog. The team then works collaboratively, holding a daily stand-up to track progress and remove blockers. At the end of the sprint, they deliver a working product increment, review it with stakeholders for feedback (sprint review), and reflect on improvements in a retrospective.

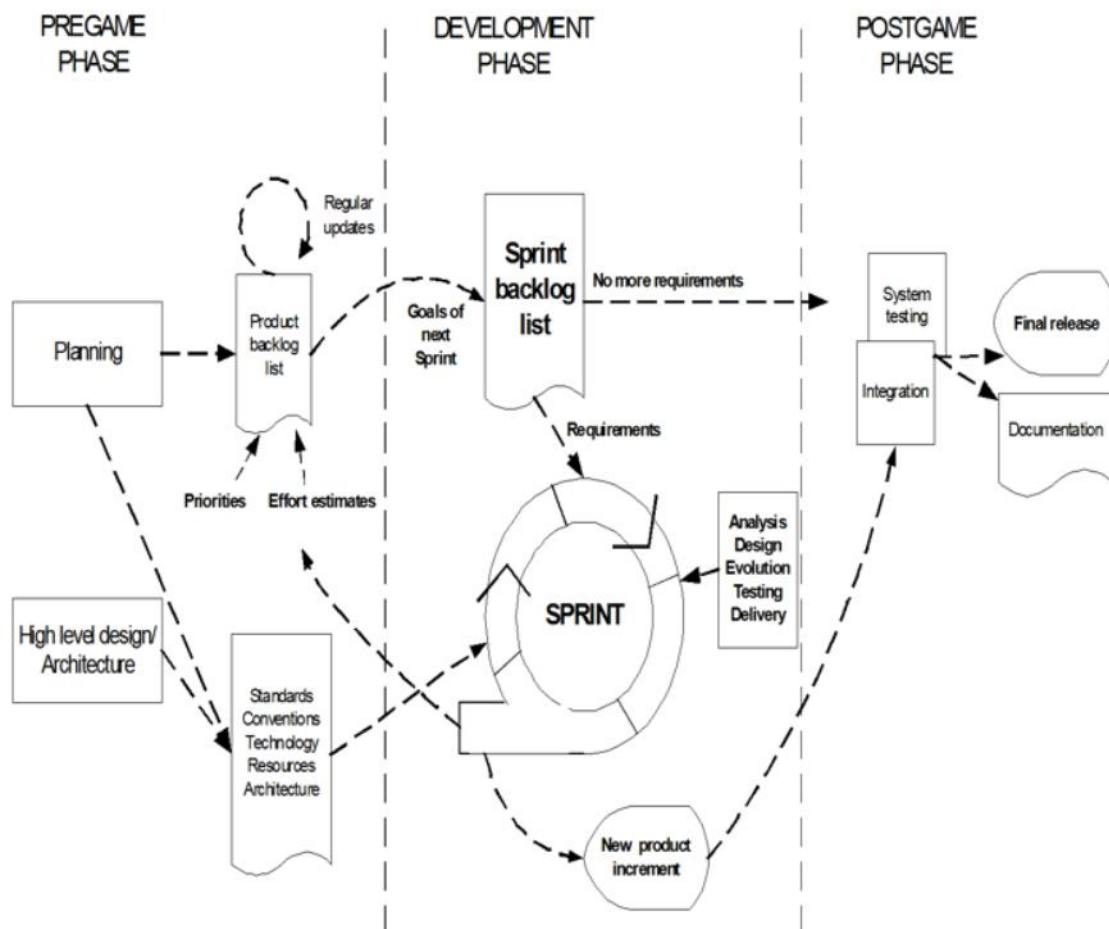


Figure: Scrum Process.

Phases of Scrum

1. Pre-game:

Planning:

- Definition of the system being developed
- A Product Backlog list is created containing all the requirements that are currently known
- The requirements are prioritized and the effort needed for their implementation is estimated
- The product Backlog list is constantly updated with new and more detailed items, as well as with more accurate estimations and new priority orders
- Planning also includes the definition of the project team, tools and other resources, risk assessment and controlling issues, training needs and verification management approval

Architecture:

- The high-level design of the system including the architecture is planned based on the current items in the Product Backlog
- In case of an enhancement to an existing system, the changes needed for implementing the Backlog items are identified along with the problems they may cause
- A design review meeting is being held to go over the proposals for the implementation and decisions are made on the basis of this review

2. Game Phase (Development)

- This phase is treated as a "black box" where the unpredictable is expected
- The system is developed in Sprints
- Sprints are iterative cycles where the functionality is developed or enhanced to produce new increments.
- Each Sprint includes the traditional phases of software development: requirements, analysis, design, evolution and delivery phases.
- One Sprint is planned to last from one week to four weeks (not exceed one month).

3. Post-game (Release)

- This phase is entered when an **agreement** has been made such as the requirements are completed.
- In this case, **no more items and issues** can be found, nor can any new ones be invented.
- The system is now ready for release and the preparation for this is done during the post-game phase, including tasks such as the integration, system testing and documentation.

Roles & Responsibilities

- **Product Owner**
 - Managed Product Backlog
 - Deciding priorities and requirements
- **Scrum Master**
 - Ensure Scrum rules are followed
 - Helps team and removes problems
- **Scrum Team (Developers)**
 - Builds the product
 - Plans and completes sprint tasks
- **Customer**
 - Gives requirements and feedback
- **Management**
 - Makes final decisions and sets goals

Reason Behind to choose this model:

The project follows the Scrum model because it best fits the nature of the system, which is feature-rich, user-driven, and continuously evolving.

1. Works in Short Sprints (2–4 Weeks)

Scrum divides development into small time-boxed iterations called sprints.

- Each sprint can focus on a group of user stories such as:
 - Sprint 1 → Authentication + Dashboard
 - Sprint 2 → Weather & Market Info
 - Sprint 3 → Disease Detection
 - Sprint 4 → Irrigation & Reminders

This helps the team deliver usable features early, instead of waiting for the full system.

2. Product Backlog for Prioritization

All 30 user stories are stored in a Product Backlog.

- High priority:
 - Weather Forecast
 - Market Prices
 - Disease Detection
- Medium priority:
 - Irrigation Guidance
 - Notifications
- Lower priority:
 - Voice Interaction
 - Offline Sync

This ensures critical farmer needs are developed first.

3. Strong Focus on Stakeholder Feedback

Scrum emphasizes continuous feedback from stakeholders:

Farmers → usability, simplicity

Advisors → accuracy of crop/disease data

Admin → system performance

After every sprint:

- A working feature is shown

- Feedback is collected
- Improvements are made in the next sprint

◆ 4. Adapts to Changing Requirements

Agriculture is dynamic:

- Weather conditions change
- Crop diseases evolve
- Market prices fluctuate

Scrum allows:

- Adding new user stories anytime
- Modifying existing features

This flexibility is not possible in rigid models like Waterfall.

◆ 5. Better Risk Management

Since development is iterative:

- Errors are found early
- Features are tested continuously
- Less chance of total project failure

◆ 6. Continuous Testing & Improvement

Each sprint includes:

- Development
- Testing
- Review
- Fewer bugs
- Reliable system for farmers

2. SOFTWARE REQUIREMENTS SPECIFICATIONS

AgriAssist is a simple mobile app designed to help farmers manage their daily activities by providing real-time weather updates, market prices, and smart irrigation support in one place. It bridges the gap for rural users with features like Voice Assistant and Offline Mode, ensuring they can diagnose crop diseases and receive expert advice even without a steady internet connection. Ultimately, it aims to modernize traditional farming to reduce crop loss and boost overall productivity.

2.1 User Story Table

<i>As a/an</i>	<i>I want to</i>	<i>So that</i>	<i>Acceptance Criteria</i>
Farmer	launch the AgriAssist app and access a simple dashboard	I can quickly find the features I need without confusion or delay.	When the farmer opens the AgriAssist app, the dashboard shall load within three seconds, showing all major features with clear labels, along with a visible registration and login button.
	create an account in the AgriAssist	I can register my information and get experience.	Farmers shall register by entering name, phone, and password. Registration succeeds only with valid inputs, and a confirmation message shall appear. Invalid entries show clear error messages.
	log in to the AgriAssist app	I can securely access my saved data and continue using the app easily.	Farmers shall log in with phone and password. Invalid details show an error. Sessions remain active until logout or thirty minutes of inactivity, with passwords stored securely.
	view the weather forecast for my area	I can plan watering and protect my crops from rain or heat.	The app shall show local weather forecasts with temperature, rainfall, and humidity for at least three days ahead, refreshing every three hours.
	check the current market prices for crops	I can decide the best time to sell my harvest for profit.	Farmers shall view daily updated crop prices from trusted sources, with comparisons for at least three nearby markets.
	receive crop care and fertilizer tips	I can improve crop health and increase yield.	The system shall display seasonal crop care and fertilizer tips

		updated weekly by experts, with an option to save them offline.
know when and how much to water my crops	I can save water and ensure proper crop growth.	The app shall give irrigation advice based on crop and weather, send watering reminders, and allow farmers to customize schedules.
Identify crop diseases using photos	I can treat them early and avoid crop loss.	Farmers can upload photos or select symptoms. The system shall suggest possible diseases with treatments or forward the query to an expert if unclear.
ask questions using my voice in my local language	I can use the app easily without needing to type or read.	Farmers shall ask questions using voice in their local language, and the app shall respond by text or audio for ease of use.
Access saved information without internet	I can still use the app in poor network areas.	Farmers shall access saved weather, tips, and records without internet. Updates sync automatically when connection is restored.
receive reminders for farming tasks	I can manage planting, watering, and harvesting on time.	The system shall allow reminders for planting, watering, and harvesting, sending timely notifications even if the app is closed.
select my crop type in the system	I can receive personalized farming advice.	Farmers shall select crop type from a list, and the system shall save it for generating recommendations.
receive irrigation notifications	I can water my crops at the right time.	The app shall send notifications based on irrigation schedules considering crop type and weather.
enter crop symptoms manually when images aren't available	I can identify diseases without uploading images.	Farmers shall input symptoms, and the system shall suggest possible diseases with treatments.
view treatment suggestions for identified diseases	I can take proper action quickly.	The system shall display treatment steps clearly after disease identification.
receive voice responses in my local language	I can understand information easily.	The app shall respond to voice queries with audio or text in the selected local language.

	store important farming data offline	I can access it without internet connection.	The system shall save selected weather, tips, and records locally and allow offline access.
	I want to view my farming history and past activities	I want to view my farming history and past activities	The system shall store and display past farming activities (such as planting, watering, and harvesting) organized by date.
	sync offline data automatically	my information remains updated.	The system shall synchronize stored data when internet connection is restored.
Farm laborer	view assigned farming tasks	I can perform my daily activities correctly.	The system shall display assigned tasks such as planting, watering, and harvesting clearly.
	I want to mark tasks as completed	I can track my finished work.	The system shall allow laborers to mark assigned tasks as completed, and the status shall be updated in the system.
	receive reminders for assigned tasks	I complete work on time.	The system shall send notifications for scheduled tasks even if the app is closed.
Agricultural advisor	answer farmers disease or fertilizer related queries	I can guide them with proper agricultural knowledge.	Agricultural advisors shall reply to farmer queries within twenty-four hours, and farmers shall be notified when responses are available.
	update and post tips on crop care, seasonal advice, and fertilizers	farmers can receive the latest and accurate farming knowledge.	Agricultural advisors shall add or update farming tips, which appear on farmer dashboards within minutes, while outdated ones are archived.
	upload disease information to the app	farmers can access accurate and helpful data for diagnosing and treating their crops.	Agricultural advisors shall upload disease data with symptoms and treatments, categorized by crop type and made available in the system.
	organize and categorize farming tips	farmers receive relevant and accurate information.	Agricultural advisors shall add and categorize tips by crop type, and the system shall display them accordingly.
System administrator	add, update, or remove user accounts	the platform remains secure and up-to-date.	System administrators shall add, update, or remove accounts, prevent duplicates, and reset passwords securely when required.

	track server usage and user activity	I can maintain smooth app performance.	System administrators shall view reports on usage and activity, with alerts triggered if server performance thresholds are exceeded.
	upload new crop disease data, weather sources, and market links	farmers get fresh and reliable content.	System administrators shall upload new disease, weather, or price sources, which update within fifteen minutes, while outdated links are flagged.
	update system data sources	farmers always receive accurate and up-to-date information.	System administrators shall update weather, market, and disease data sources, and updates shall reflect within a short time.

2.2 Requirements Traceability Matrix

2.2.1 Functional Requirements

NO.	Requirement Description
1.0	Farmer home page – All features availability
1.1	Registration
1.1.1	Taking full name
1.1.2	Taking email address
1.1.3	Taking username
1.1.4	Taking password
1.1.5	Taking confirm password
1.1.6	Taking phone number
1.1.7	Taking location (for weather & market info)
1.1.8	Submitting the registration form
1.1.9	Resetting the form
1.2	Login
1.2.1	Taking username
1.2.2	Taking password
1.2.3	Providing “Forgot Password” option
1.2.3.1	Submitting email address to recover account [ref 1.2.3]
1.2.4	Submitting login [ref 1.2]

1.3	Dashboard
1.3.1	Side menu with navigation
1.3.2	Dashboard button [ref 1.3]
1.3.3	Weather forecast widget
1.3.4	Market prices widget
1.3.5	Crop care & tips widget
1.3.6	View profile button
1.3.6.1	Show name
1.3.6.2	Show email
1.3.6.3	Show gender
1.3.6.4	Show phone number
1.3.6.5	Show date of birth
1.3.6.6	Show profile picture
1.4	Smart Irrigation Support
1.4.1	Selecting crop type
1.4.2	Viewing recommended watering schedule
1.4.3	Receiving irrigation notifications
1.5	Crop Disease Finder
1.5.1	Uploading photo of crop
1.5.2	Entering symptoms manually
1.5.3	Receiving disease diagnosis
1.5.4	Viewing suggested treatments
1.6	Voice Assistant
1.6.1	Accepting farmer's voice input
1.6.2	Responding in local language
1.7	Offline Mode
1.7.1	Saving weather updates locally
1.7.2	Saving farming tips & records locally
1.7.3	Accessing offline data without internet
1.8	Farming Calendar
1.8.1	Adding planting schedule
1.8.2	Adding watering schedule
1.8.3	Adding fertilizing schedule
1.8.4	Adding harvesting schedule
1.8.5	Receiving reminders & alerts

2.0	Farm laborer home page – All features availability
2.1	Registration [ref 1.1]
2.2	Login [ref 1.2]
2.3	Dashboard
2.3.1	Side menu with navigation
2.3.2	Dashboard button [ref 2.3]
2.3.3	Assigned tasks widget
2.3.4	Completed tasks widget
2.3.5	Scheduled tasks widget
2.3.6	View profile button [ref 1.3.6]
3.0	Agricultural advisor home page – All features availability
3.1	Registration
3.1.1	Taking full name
3.1.2	Taking email address
3.1.3	Taking username
3.1.4	Taking password
3.1.5	Taking confirm password
3.1.6	Taking phone number
3.1.7	Taking location (for weather & market info)
3.1.8	Taking valid documents for identification
3.1.9	Submitting the registration form
3.1.10	Resetting the form
3.2	Login [ref 1.2]
3.3	Dashboard
3.3.1	Side menu with navigation
3.3.2	Dashboard button [ref 1.3]
3.3.3	Weather forecast widget
3.3.4	Market prices widget
3.3.4	Crop care & tips widget
3.3.5	View profile button
3.3.5.1	Show name
3.3.5.2	Show email
3.3.5.3	Show gender
3.3.5.4	Show phone number
3.3.5.5	Show date of birth

3.3.5.6	Show profile picture
3.3.5.7	Show documents
3.4	Agricultural advisor functionalities
3.4.1	Answer farmer queries (text/voice)
3.4.2	Upload crop care tips
3.4.3	Upload seasonal advice & fertilizer info
3.4.4	Upload new disease information with solutions
4.0	System administrator home page – All features availability
4.1	Login [ref 1.2]
4.2	Dashboard [ref 1.3]
4.3	System administrator functionalities
4.3.1	Add, update, or remove user accounts
4.3.2	Track server usage and user activity
4.3.3	Upload new crop disease data
4.3.4	Update weather data sources
4.3.5	Update market price data sources

2.2.2 Non-Functional Requirements

2.1	Performance
2.1.1	The app dashboard shall load within 2 seconds of launch.
2.1.2	Weather data shall refresh every hour without manual intervention.
2.1.3	Market price updates shall reflect within 15 minutes of data source changes.
2.2	Reliability
2.2.1	System uptime shall be maintained at 99% especially during peak agricultural seasons.
2.2.2	Offline-stored data shall remain accessible without data loss during connectivity failure.
2.2.3	Data synchronization shall complete successfully within 60 seconds of internet restoration.
2.3	Scalability
2.3.1	The system shall support at least 1,000 concurrent users without degradation in response time.
2.3.2	The server architecture shall allow horizontal scaling to accommodate future user growth.
2.3.3	Ability to add new features and data sources shall be seamless without system downtime.
2.4	Security
2.4.1	User passwords shall be stored using industry-standard hashing (e.g., bcrypt).
2.4.2	All user data shall be transmitted over encrypted HTTPS connections.
2.4.3	Sessions shall expire after 30 minutes of inactivity and users shall be required to log in again.
2.4.4	Duplicate account creation shall be prevented by the system.
2.5	Usability
2.5.1	The interface shall use simple, icon-based navigation with clear labels suitable for farmers with limited literacy.
2.5.2	All major features shall be accessible within 2 taps from the dashboard.
2.5.3	The app shall support Bangla as the primary language for all labels, tips, and notifications.
2.6	Availability
2.6.1	The app shall be available 24/7 for all registered users.
2.6.2	Core offline features (saved weather data, farming tips, records) shall remain functional without internet.
2.6.3	Push notifications for farming reminders shall fire even when the app is running in the background or closed.
2.7	Maintainability
2.7.1	System administrators shall be able to update data sources without causing system downtime.
2.7.2	New crop disease data, weather sources, or market links shall reflect within 15 minutes of upload.

2.7.3	Outdated data links and disease entries shall be automatically flagged for administrative review.
2.8	Privacy
2.8.1	Personal user data (name, phone number, location) shall not be shared with third parties without user consent.
2.8.2	Agricultural advisors' identity documents shall be stored securely and accessible only by system administrators.

3. SOFTWARE DESIGN

3.1 System Design

3.1.1 Use Case Diagram

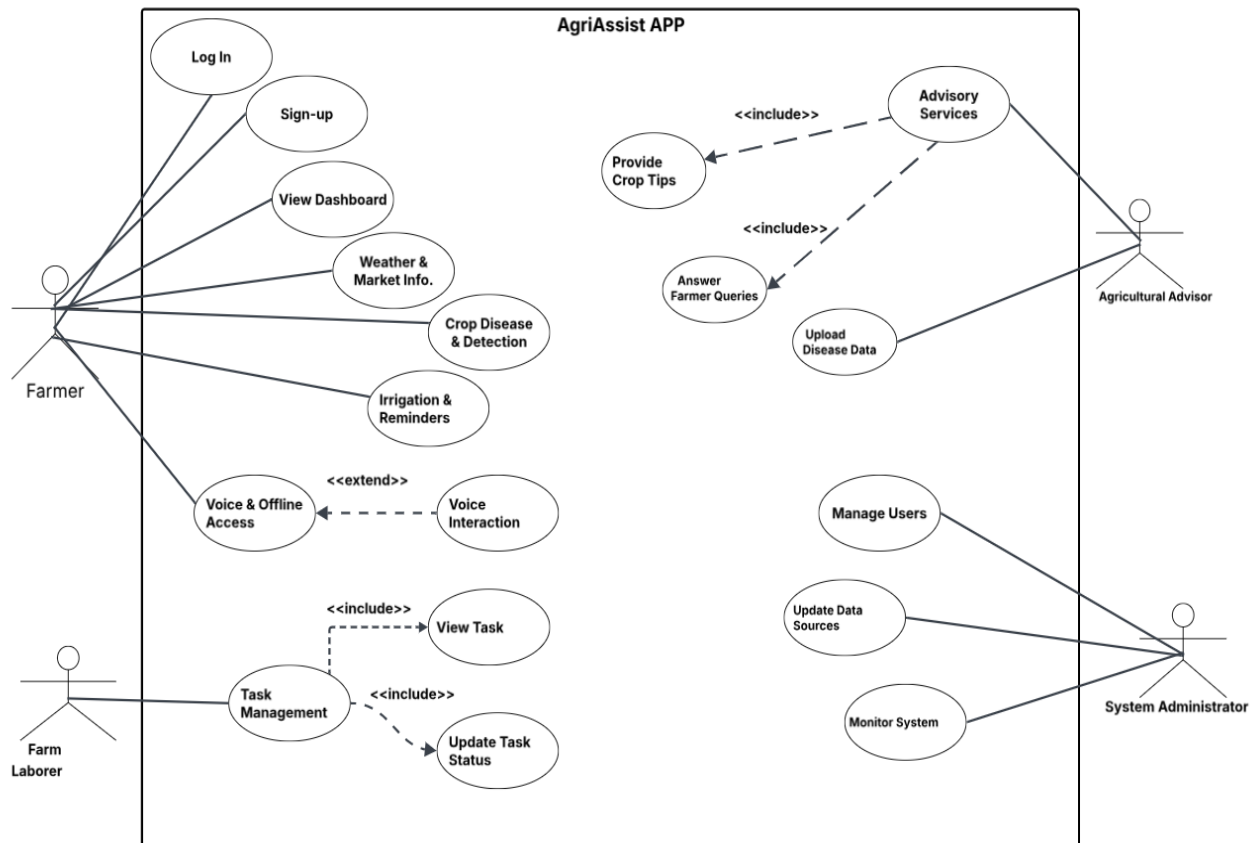


Figure: Use Case Diagram

3.1.2 Class Diagram

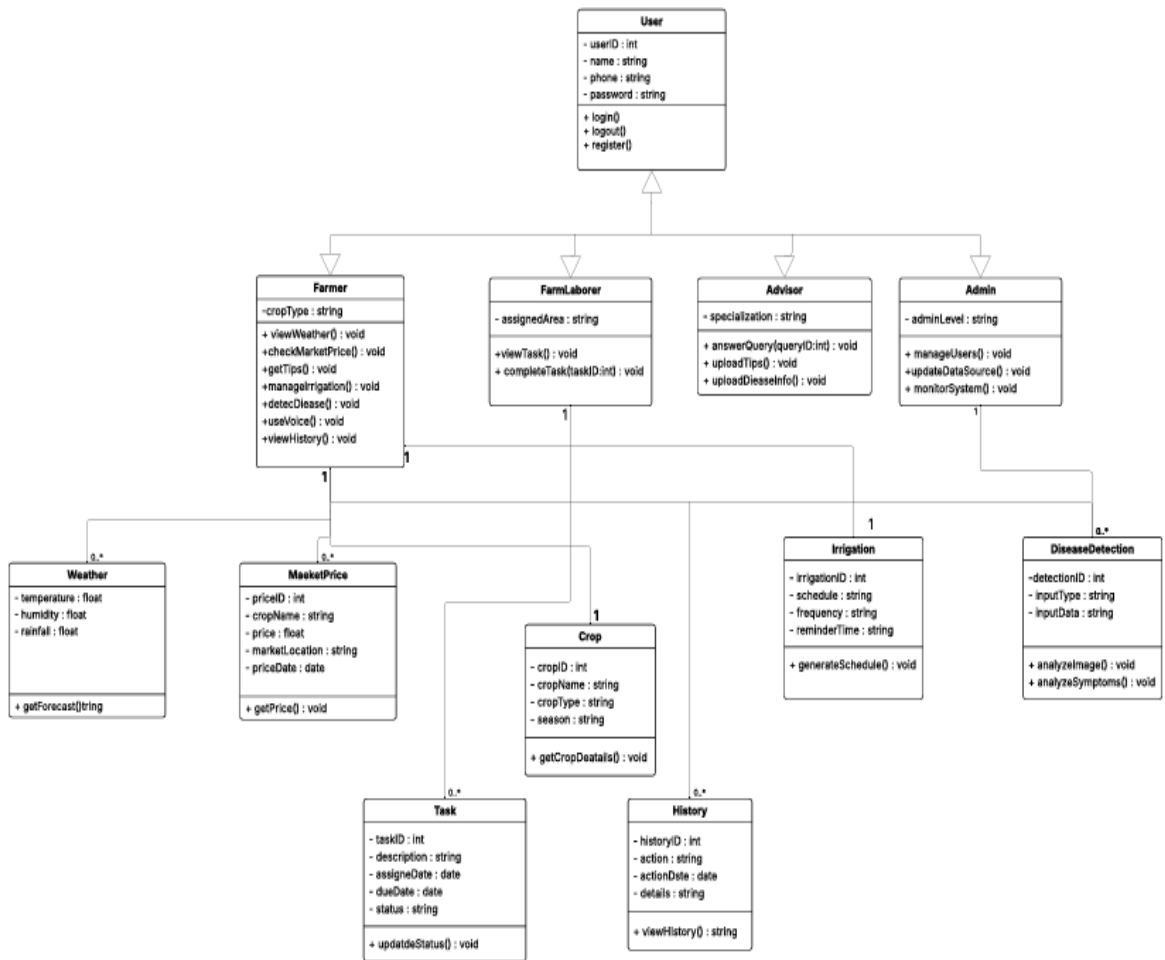


Figure: Class Diagram

3.2 Wireframe Design

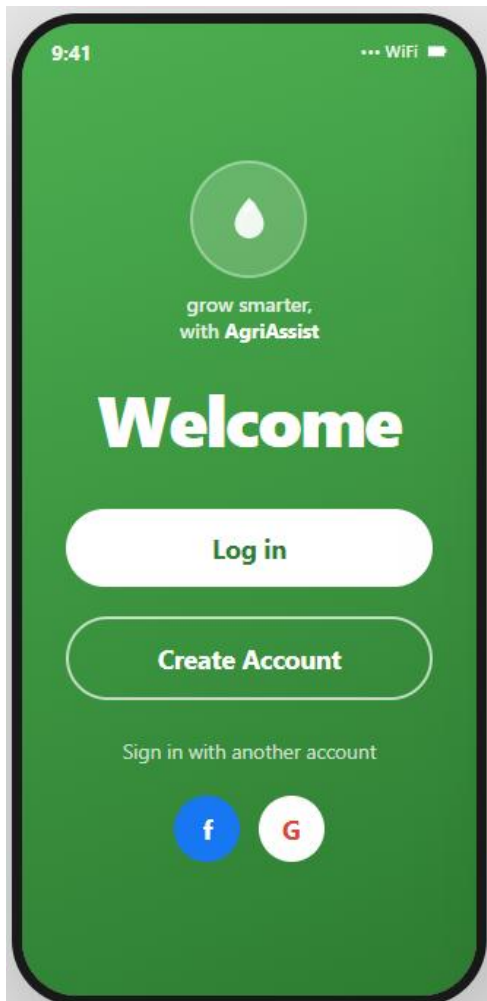


Figure: Home

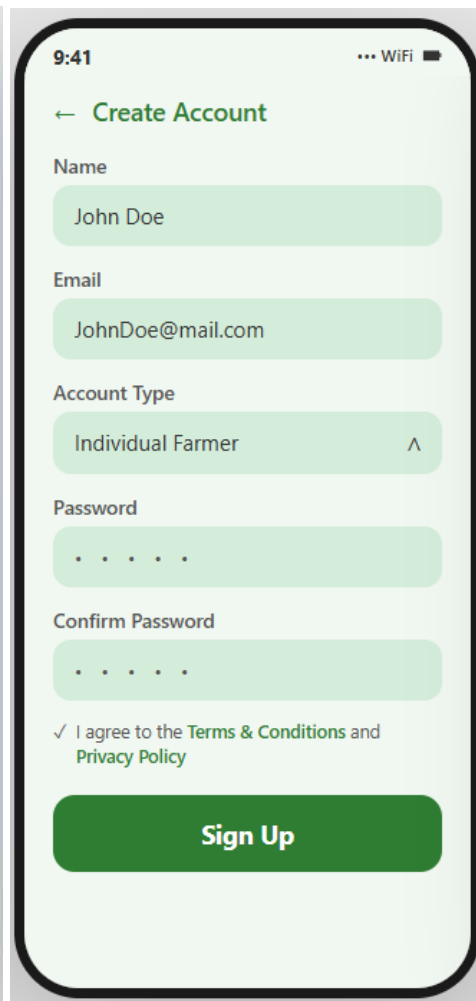


Figure: Registration

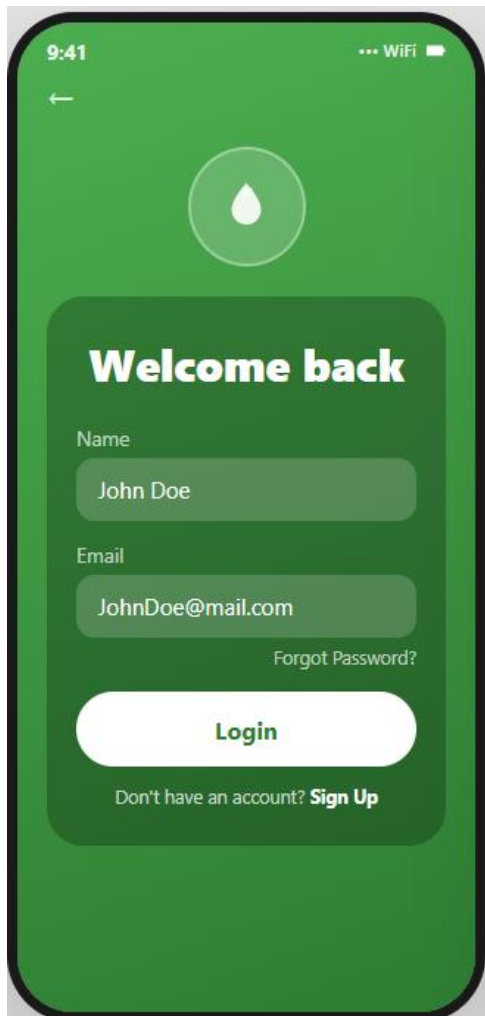


Figure: Login

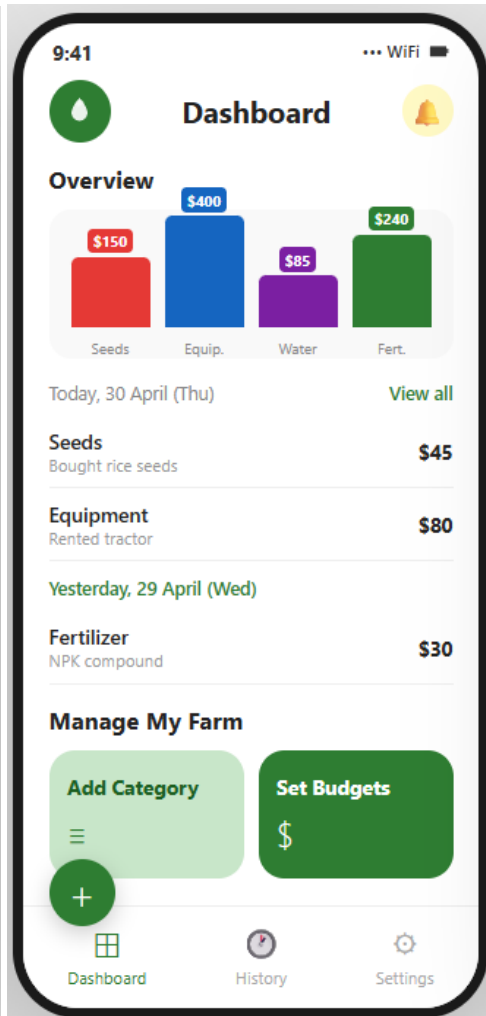


Figure: Dashboard

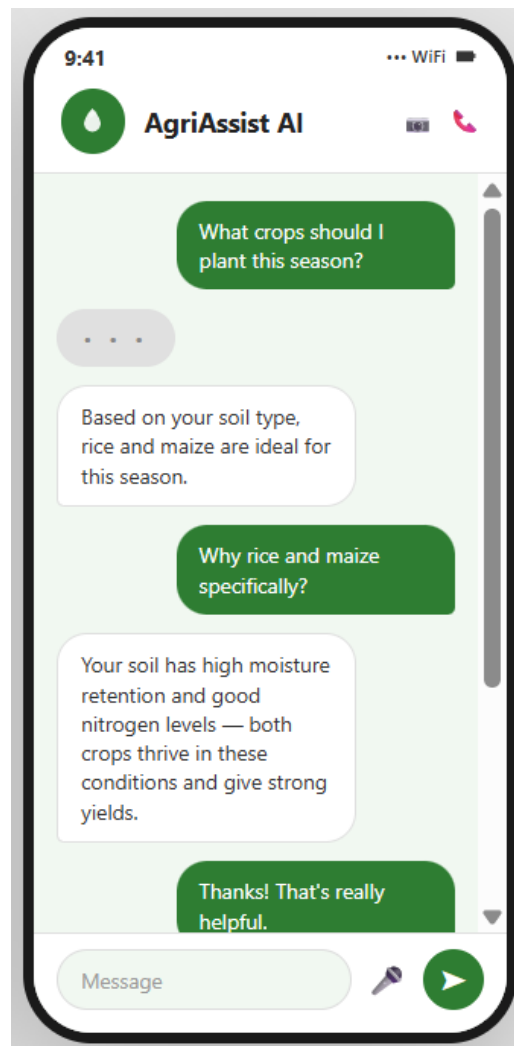


Figure: AgriAssist Chat

4. GIT WORKFLOW

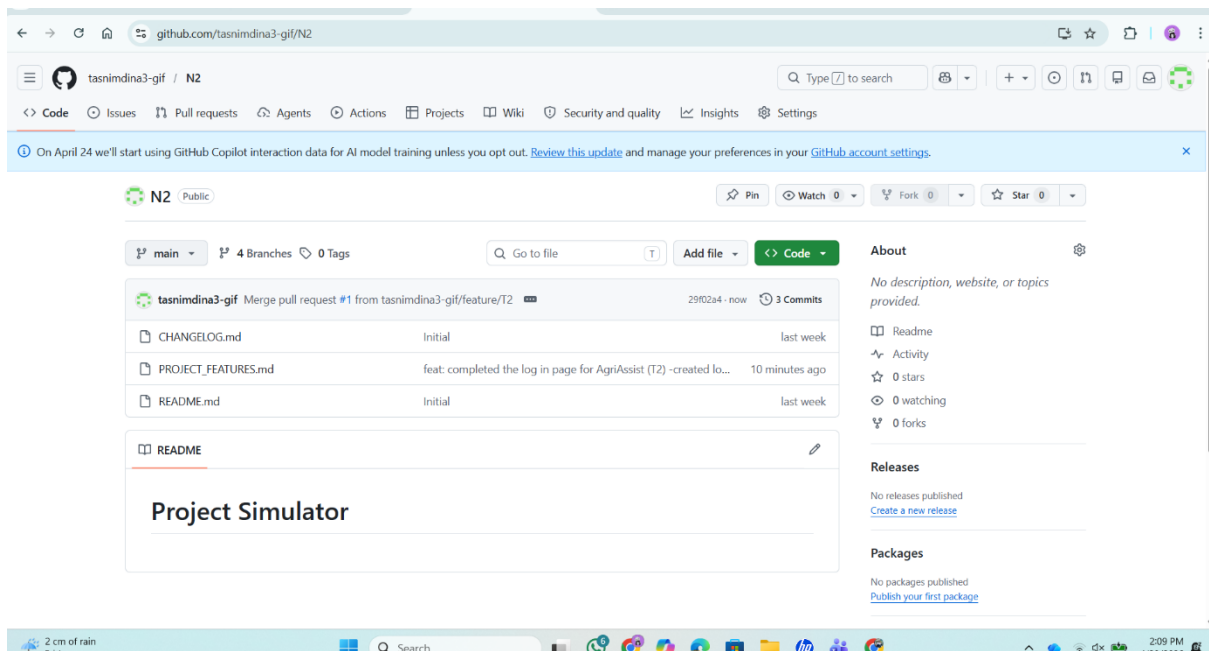


Figure: GitHub mother repository (main branch)

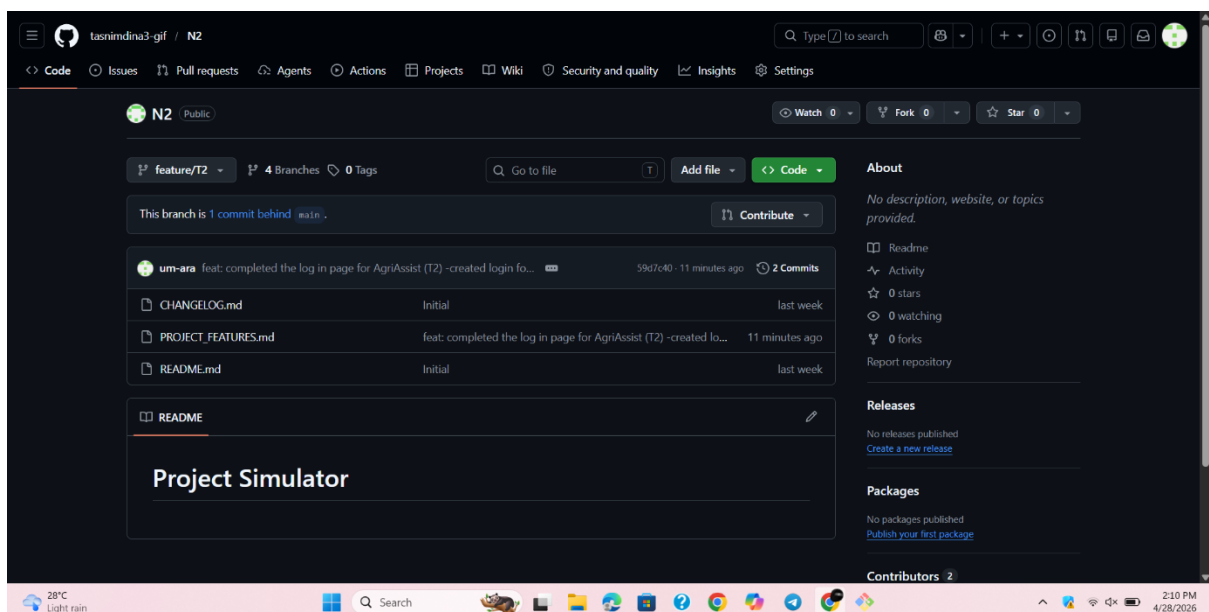


Figure: GitHub clone repository (feature/T2 branch)

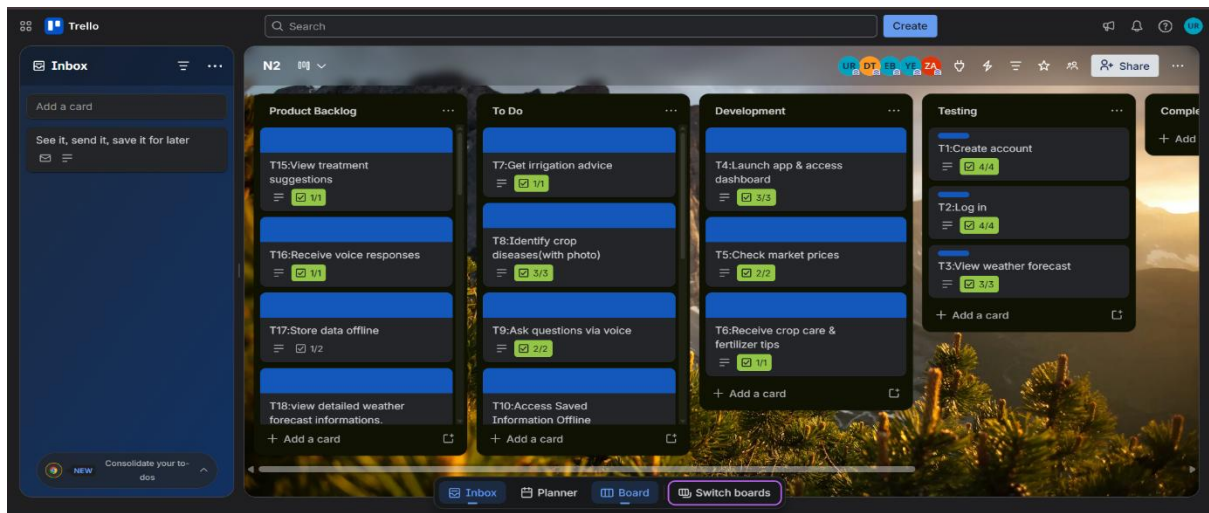


Figure: Trello board initial stage

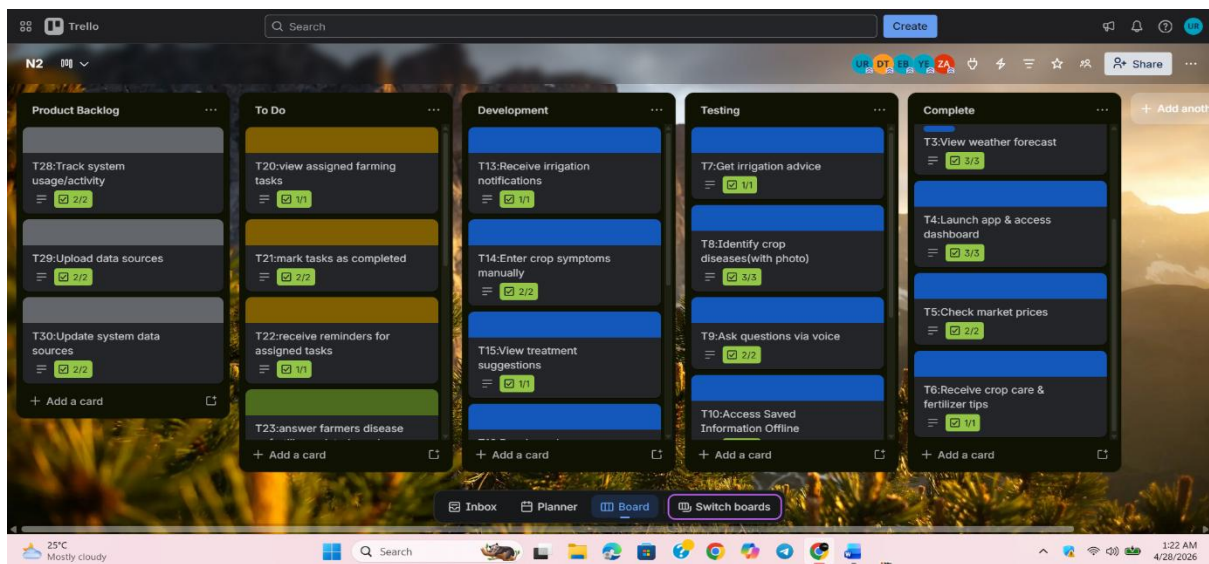


Figure: Trello board intermediate stage

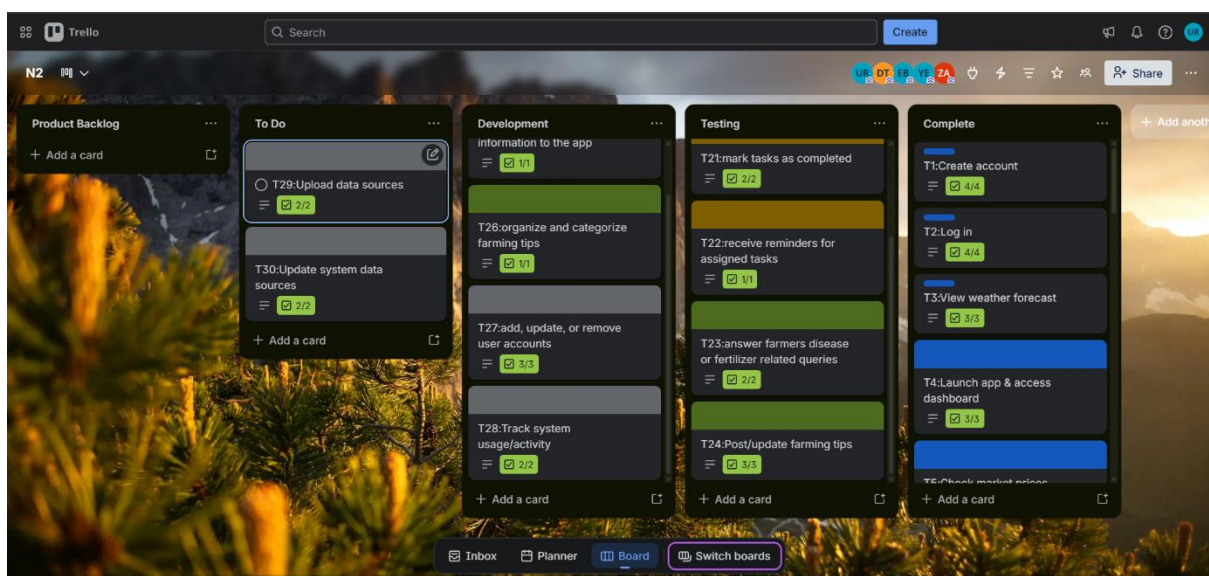


Figure: Trello board end stage

5. SOFTWARE TESTING

Project Name: Agri-Assist		Test Designed by: Emon Biswas		
Test Case ID: TC_01		Test Designed date: 01 May ,2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Offline Storage		Test Execution date:		
Test Title: Verify offline storage and accessibility of farming data				
Description: Verify local data storage and offline accessibility.				
Precondition: The user is logged into the application and has initial data synced.				
Dependencies: Local database				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Trigger local save for weather, tips, and records. 2. Disable internet connection 3. Access saved weather, tips, and records while offline.	Data: Weather/Tips, Stored files Status: Offline	Application remains operational in offline mode. Local storage is accessible and displays data without internet.		
Post Condition: Important farming data is securely stored on the device and remains fully accessible to the user during connectivity outages.				

Project Name: Agri-Assist		Test Designed by: Emon Biswas		
Test Case ID: TC_02		Test Designed date: 01 May ,2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: View updated market prices		Test Execution date:		
Test Title: Verify daily crop price updates and market comparisons.				
Description: Test the ability for farmers to view current crop prices and compare them across nearby markets.				
Precondition: The user is logged into the application, and location services are active.				
Dependencies: Integration with a real-time market price database.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Market Prices section. 2. Select "Compare Markets" option. 3. Verify price update.	Crop: Rice/Jute Radius: 40km Current Date	Daily updated crop prices are displayed. System shows price comparisons for at least three nearby markets. The price shown reflects the current day's update.		
Post Condition: The user can view accurate market data to decide the best time to sell their harvest for profit.				

Project Name: Agri-Assist		Test Designed by: Umara Rashid		
Test Case ID: TC_03		Test Designed date: 02 May, 2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: View Weather Forecast		Test Execution date:		
Test Title: Verify local weather forecast display and refresh cycle				
Description: Test the ability for farmers to view local weather forecasts with temperature, rainfall, and humidity for at least 3 days ahead.				
Precondition: The user is logged into the application and location services are enabled.				
Dependencies: Integration with a real-time weather data API.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to Weather Forecast section from the dashboard. 2. Observe forecast data displayed on screen. 3. Wait 3 hours and verify automatic data refresh.	Location: Dhaka Current Date Elapsed time: 3 hours	Weather forecast loads within 3 seconds. Temperature, rainfall, and humidity displayed for at least 3 days ahead. Forecast data refreshes automatically every 3 hours.		
Post Condition: The farmer can view an up-to-date 3-day weather forecast to plan crop watering and protection activities effectively.				

Project Name: Agri-Assist		Test Designed by: Umara Rashid		
Test Case ID: TC_04		Test Designed date: 02 May, 2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Crop Disease Finder		Test Execution date:		
Test Title: Verify crop disease identification via photo upload and treatment suggestion				
Description: Test the ability for farmers to upload a photo of their crop to receive a disease diagnosis and actionable treatment suggestions.				
Precondition: The user is logged into the application and camera/gallery permissions are granted.				
Dependencies: Disease detection model or expert system backend; agricultural advisor escalation module.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Crop Disease Finder section. 2. Upload a photo of a diseased crop. 3. View diagnosis result and treatment suggestions.	Crop: Rice Image: rice_yellowing.jpg Crop Type: Rice	Crop Disease Finder loads with upload and symptom options. Photo is accepted and processed. System displays disease name with treatment steps; if unclear, query is forwarded to an agricultural advisor.		
Post Condition: The farmer receives a disease diagnosis with actionable treatment suggestions, enabling early intervention to prevent crop loss.				

Project Name: Agri-Assist		Test Designed by: Yasin Ali Emon		
Test Case ID: TC_05		Test Designed date: 01 May ,2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Upload disease information to the app		Test Execution date:		
Test Title: Verify uploading disease information to the app				
Description: Test the functionality of uploading crop disease data like symptoms and treatments into the system.				
Precondition: User is logged in as Agricultural Advisor or Admin.				
Dependencies:				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the dashboard 2. Navigate to “Disease Information” section 3. Click on “Add Disease Info” or Upload option 4. Enter disease name 5. Enter symptoms and treatments 6. Select crop type. 7. Click Submit	Disease Name: Leaf Blight Symptoms: Yellowing leaves, brown spots Treatment: Use fungicide spray Crop Type: Rice	Agricultural advisors shall upload disease data with symptoms and treatments, categorized by crop type and made available in the system.		
Post Condition: Disease information is stored in the database and available for farmers to view.				

Project Name:Agri-Assist		Test Designed by:Yasin Ali Emon		
Test Case ID:TC_06		Test Designed date: 01 May ,2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Create Account		Test Execution date:		
Test Title:Verify Farmer Account Creation				
Description: Test the functionality of creating a new farmer account in the system.				
Precondition:				
Dependencies: Internet connection available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Agri-Assist app 2. Click Register / Create Account 3. Enter farmer information 4. Enter email and password 5. Click Submit / Register	Name: Rahim Uddin Email: rahim@gmail.com Account Type: Individual Farmer Password: Rahim123@ Confirm Password: Rahim123@	Farmers shall register by entering name, phone, and password. Registration succeeds only with valid inputs, and a confirmation message shall appear. Invalid entries show clear error messages.		
Post Condition: Farmer account information is stored in the database and available for login.				

Project Name: Agri-Assist		Test Designed by: Dina Akter		
Test Case ID: TC_07		Test Designed date: 01 May ,2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Crop Care & Fertilizer Tips		Test Execution date:		
Test Title: Verify that seasonal crop care and fertilizer tips are displayed to farmers				
Description: Test the functionality of displaying crop care and fertilizer tips to farmers, including expert-update weekly tips and offline save option.				
Precondition: User is logged in as Farmer				
Dependencies:				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Log in as Farmer 2.Navigate to ‘Crop Care & Tips ’section 3.View the list of seasonal crop care and fertilizer tips 4.Check if tips are updated weekly by experts 5.Save a tip for offline access	Crop Type: Rice Fertilizer Tip: Apply Urea at tillering stage Tip Category: Seasonal Fertilizer	The system shall display seasonal crop care and fertilizer tips updated weekly by experts, with an option to save them offline.		
Post Condition: Farmer account information is stored in the database and available for login.				

Project Name: Agri-Assist		Test Designed by: Dina Akter		
Test Case ID: TC_08		Test Designed date: 01 May ,2026		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Access saved Information Offline		Test Execution date:		
Test Title: Verify accessing saved information without internet connection				
Description: Test the functionality of accessing saved weather, tips and records without internet connection and verify data syncs when connection is restored.				
Precondition: User is logged in as Farmer				
Dependencies: Local database				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Go to the dashboard 2.Navigate to "Offline Mode" section 3.Access saved weather, tips and records without internet 4.Restore internet connection 5.Verify updates sync automatically	Saved Weather: 3- days forecast Saved Tips: Rice fertilizer tip Saved Record: Planting date log	Farmers shall access saved weather, tips, and records without internet. Updates sync automatically when the connection is restored.		
Post Condition:				

6. CONCLUSION

AgriAssist is a simple mobile application designed to support farmers in their daily farming activities more efficiently. The system brings together important features like weather updates, market prices, crop disease detection, irrigation support, reminders, and offline access in one platform. It was developed using the Scrum model, which helped the team improve the system step by step through continuous feedback and testing. The application is designed for farmers, farm laborers, agricultural advisors, and administrators so they can work together more easily. Overall, AgriAssist shows how technology can make farming smarter, reduce crop loss, and help farmers make better decisions in their everyday work.